

Root locus tech

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Mechatronics 3DX4

Slides 8: Root Locus Techniques

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Material based on *Control Systems Engineering* by N. Nise.

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Introduction

- ▶ The **root locus** technique shows graphically how the closed-loop poles change as a system parameter is varied.
- ▶ Used to analyze and design systems for stability and transient response.
- ▶ Shows immediately the effect of variations of system parameters.

- Shows graphically the effect of varying the gain on things like percent overshoot, and settling time.
- Also shows graphically how stable a system is; shows ranges of stability, instability, and when system will start oscillating.

The Control System Problem

- The poles of the open-loop transfer function are typically easy to find and do not depend on the gain, K .
- It is thus easy to determine stability and transient response for an open-loop system.
- Let $G(s) = \frac{N_G(s)}{D_G(s)}$ and $H(s) = \frac{N_H(s)}{D_H(s)}$.

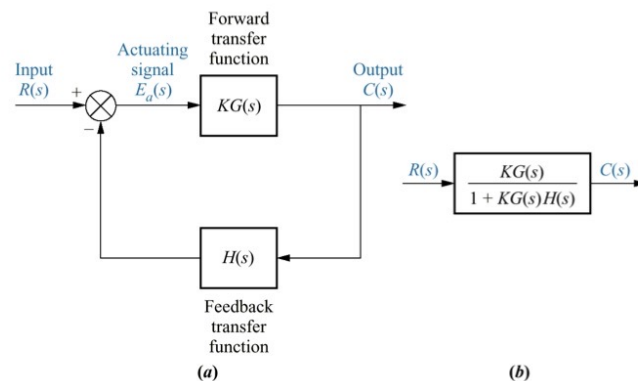


Figure 8.1.

The Control System Problem - II

- Our closed transfer function is thus

$$K \frac{N_G(s)}{D_G(s)}$$

$$T(s) = \frac{1}{1 + K \frac{N_G(s) N_H(s)}{D_G(s) D_H(s)}} \quad (1)$$

$$= \frac{K N_G(s) D_H(s)}{D_G(s) D_H(s) + K N_G(s) N_H(s)} \quad (2)$$

- We thus see that we have to factor the denominator of $T(s)$ to find the closed-loop poles, and they will be a function of K .

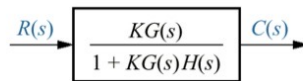


Figure 8.1(b).

The Control System Problem - III

- For example, if $G(s) = \frac{s+1}{s(s+2)}$ and $H(s) = \frac{s+3}{s+4}$, our closed-loop transfer function is:

$$T(s) = \frac{K(s+1)(s+4)}{s(s+2)(s+4) + K(s+1)(s+3)} \quad (3)$$

$$= \frac{K(s+1)(s+4)}{s^3 + (6+K)s^2 + (8+4K)s + 3K} \quad (4)$$

- To find the poles, we would have to factor the polynomial for a specific value of K .
- The root-locus will give us a picture of how the poles will vary with K .

K是这个系统的增益，对于open loop system，其不会影响pole，但是闭环系统则会影响pole的位置。当K趋近于无穷大时，pole的位置会趋近于开环系统的零点位置。

系统具K会影响pole的位置，变得难分析。

Vector Representation of Complex Numbers

- ▶ Any complex number, $\sigma + j\omega$, can be represented as a vector.
- ▶ It can be represented in polar form with magnitude M , and an angle θ , as $M\angle\theta$.
- ▶ If $F(s)$ is a complex function, setting $s = \sigma + j\omega$ produces a complex number. For $F(s) = (s + a)$, we would get $(\sigma + a) + j\omega$.

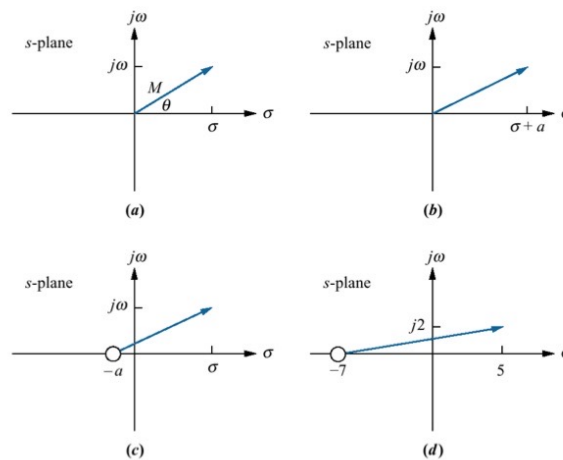
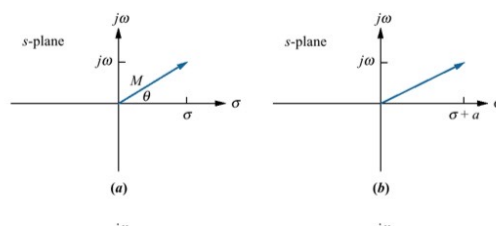


Figure 8.2.

Vector Representation of Complex Numbers - II

- ▶ If we note that function $F(s) = (s + a)$ has a zero at $s = -a$, we can alternately represent $F(\sigma + j\omega)$ as originating at $s = -a$, and terminating at $\sigma + j\omega$.
- ▶ To multiply and divide the polar form complex numbers, $z_1 = M_1\angle\theta_1$ and $z_2 = M_2\angle\theta_2$, we get

$$z_1 z_2 = M_1 M_2 \angle(\theta_1 + \theta_2) \quad \frac{z_1}{z_2} = \frac{M_1}{M_2} \angle(\theta_1 - \theta_2) \quad (5)$$



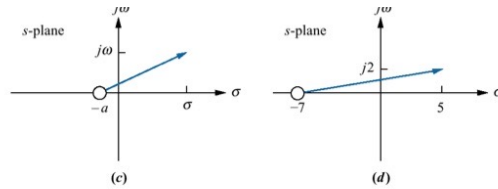


Figure 8.2.

Polar Form and Transfer Functions

- For a transfer function, we have:

$$G(s) = \frac{(s + z_1) \cdots (s + z_m)}{(s + p_1) \cdots (s + p_n)} = \frac{\prod_{i=1}^m (s + z_i)}{\prod_{i=1}^n (s + p_i)} = M_G \angle \theta_G \quad (6)$$

where

$$M_G = \frac{\prod_{i=1}^m |(s + z_i)|}{\prod_{i=1}^n |(s + p_i)|} = \frac{\prod_{i=1}^m M_{z_i}}{\prod_{i=1}^n M_{p_i}} \quad (7)$$

and

$$\theta_G = \Sigma \text{zero angles} - \Sigma \text{pole angles} \quad (8)$$

$$= \Sigma_{i=1}^m \angle(s + z_i) - \Sigma_{j=1}^n \angle(s + p_j) \quad (9)$$

这里引入了用复数向量方法来表示传递函数G。

Polar Form and Transfer Functions eg.

- Use Equation 6 to evaluate $F(s) = \frac{(s + 1)}{s(s + 2)}$ at $s = -3 + j4$.

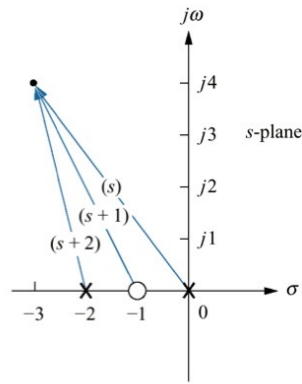


Figure 8.3.

$$F(-3+4j) = \frac{-2+4j}{(-3+4j)(4j-1)}$$

$$\angle \theta_1 = \pi - \tan^{-1}\left(\frac{4}{2}\right) = 2.03 \text{ rad}$$

$$-2+4j = 2\sqrt{5} e^{2.03j}$$

$$-3+4j : \angle \theta_2 : -\arctan\left(\frac{4}{3}\right) + \pi = 2.21 \text{ rad}$$

$$= 5 e^{2.21j}$$

$$4j-1 : \angle \theta_3 = \pi - \arctan\left(\frac{4}{1}\right) = 1.82 \text{ rad}$$

$$= \sqrt{17} e^{1.82j}$$

$$x = 0.217 \times \cos(-2) = -0.09$$

$$y = 0.217 \times \sin(-2) = -0.197$$

$$\therefore F = \frac{2\sqrt{5} e^{2.03j}}{5\sqrt{17} e^{2.21+1.82j}} = 0.217 e^{-2j} = -0.09 - 0.197j$$

Polar Form and Transfer Functions eg.

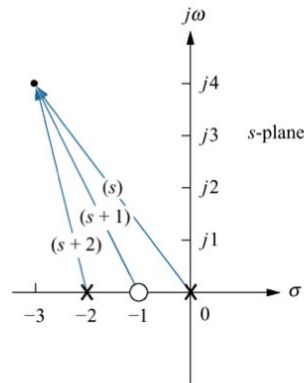


Figure 8.3.

Check answer using matlab:

```

1 s=tf('s')
2 F=(s+1)/(s*(s+2))
3 s1=evalfr(F,-3+j*4)
4 M=abs(s1)
5 theta=angle(s1)
6 theta*180/pi

```

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Root Locus Introduction

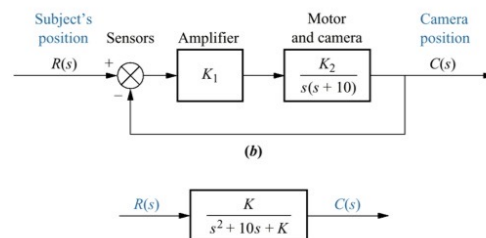
- System below can automatically track subject wearing infrared sensors.
- Solving for the poles using the quadratic equation, we can create the table below for different values of K .

Table 8.1.

K	Pole 1	Pole 2
0	-10	0
5	-9.47	-0.53
10	-8.87	-1.13
15	-8.16	-1.84
20	-7.24	-2.76
25	-5	-5
30	$-5 + j2.24$	$-5 - j2.24$
35	$-5 + j3.16$	$-5 - j3.16$



(a)



(b)

40	$-5 + j3.87$	$-5 - j3.87$
45	$-5 + j4.47$	$-5 - j4.47$
50	$-5 + j5$	$-5 - j5$

where $K = K_1 K_2$
(e)

Figure: 8.4

Root Locus Introduction - II

- We can plot the poles from Table 8.1. labelled by their corresponding gain.

Table 8.1.

K	Pole 1	Pole 2
0	-10	0
5	-9.47	-0.53
10	-8.87	-1.13
15	-8.16	-1.84
20	-7.24	-2.76
25	-5	-5
30	$-5 + j2.24$	$-5 - j2.24$
35	$-5 + j3.16$	$-5 - j3.16$
40	$-5 + j3.87$	$-5 - j3.87$
45	$-5 + j4.47$	$-5 - j4.47$
50	$-5 + j5$	$-5 - j5$

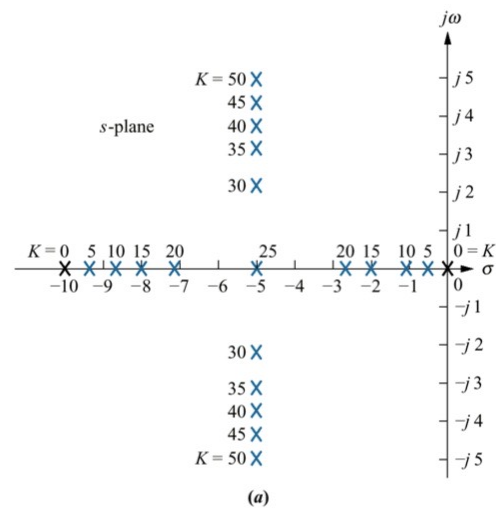


Figure: 8.5

Root Locus Introduction - III

- We can go a step further, and replace the individual poles with their paths.
- We refer to this graphical representation of the path of the poles as we vary the gain, as the **root locus**.
- We **will focus** our discussion on **$K \geq 0$** .
- For pole $\sigma_D + j\omega_D$, $T_s = \frac{4}{\sigma_D}$, $T_p = \frac{\pi}{\omega_D}$, and $\zeta = \frac{|\sigma_D|}{\omega_n}$.

$j\omega$
↑

↑

$j\omega$
↑

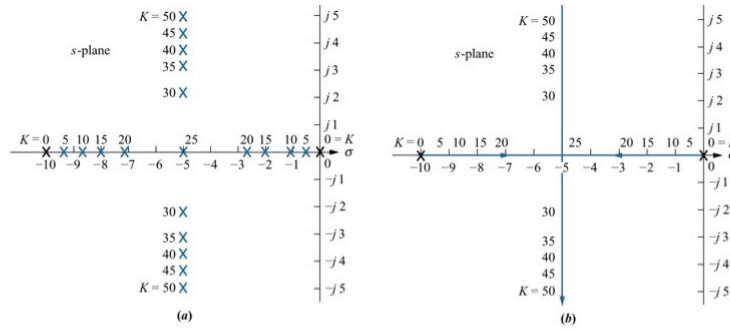
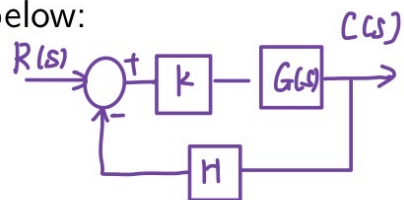


Figure 8.5.

Root Locus Properties

- ▶ For second-order systems, we can easily factor a system and draw the root locus.
- ▶ We do not want to have to factor for higher-order systems (5th, 10th etc.) for multiple values of K !
- ▶ We will develop properties of the root locus that will allow us to rapidly **sketch** the root locus of higher-order systems.
- ▶ Consider the closed-loop transfer function below:

$$T(s) = \frac{KG(s)}{1 + KG(s)H(s)}$$



- ▶ A pole of $T(s)$ exists when

$$KG(s)H(s) = -1 = 1\angle(2k + 1)180^\circ \quad k = 0, \pm 1, \pm 2, \dots \quad (10)$$



这里的T是我们标准的closed feedback loop transfer function, 所以我们让其分母等于0找出pole, 来画root locus。

Root Locus Properties - II

- Equation 10 is equivalent to

$$|KG(s)H(s)| = 1 \quad (11)$$

and

$$\angle KG(s)H(s) = (2k + 1)180^\circ \quad (12)$$

- Equation 12 says that any s' that makes the angle of $KG(s)H(s)$ be an odd multiple of 180° is a pole for some value of K .
- Given s' above, the value of K that s' is a pole of $T(s)$ for is found from Equation 11 as follows:

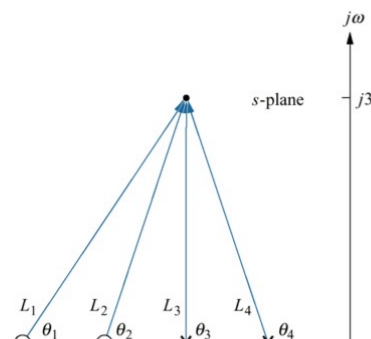
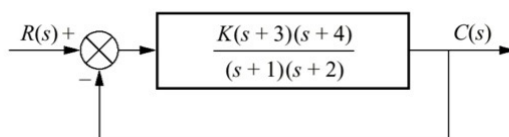
$$K = \frac{1}{|G(s)||H(s)|} \quad (13)$$

变成数学条件就是： $1 + KG(s)H(s) = 0$ ，即 $KG(s)H(s) = -1$ ，变成幅值和角度关系就是 $|KG(s)H(s)| = 1$ 以及 $\angle = (2k+1)*180$

Root Locus Properties eg.

- For system below, consider $s = -2 + j3$ and $s = -2 + j(\sqrt{2}/2)$.

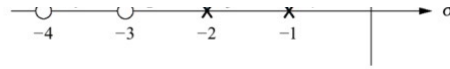
$$K = \frac{1}{|G(s)||H(s)|} = \frac{1}{3}$$



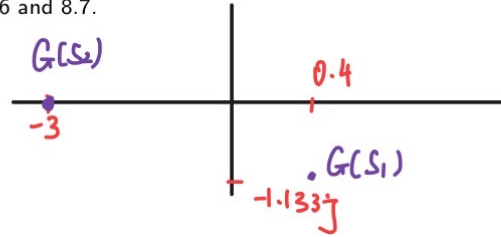
$\angle = (2k+1)*180$

$$G(s_1) = 0.4 - 1.133j$$

$$G(s_2) = -3 - 0j$$



Figures 8.6 and 8.7.



Sketching Root Locus

► Now give a set of rules so that we can quickly sketch a root locus, and then we can calculate exactly just those points of particular interest.

1. **Number of branches:** a **branch** is the path a single pole traverses. *The number of branches thus equals the number of poles.*
2. **Symmetry:** As complex poles occur in conjugate pairs, a *root locus must be symmetric about the real axis.*

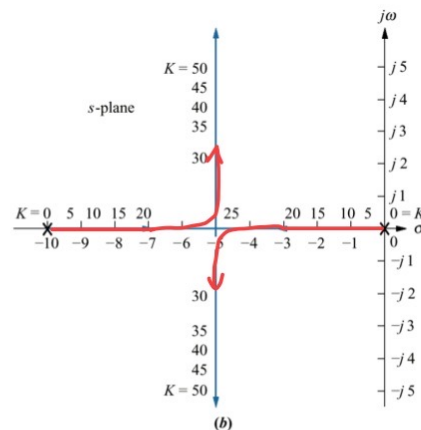


Figure 8.5.

Root locus画图怪谈:

1. **Branch 就是root移动的路径**, 有几个branch取决于有几个pole。Root locus从pole出发, 回到zero。
2. 如果出现complex root, 他们肯定是conjugate的, 即 $\sigma \pm j\omega$, 所以 root locus肯定关于real axis对称。

Sketching Root Locus - II

3. Real-axis segments: For $K > 0$, the root locus only exists on the real axis to the left of an odd number of finite open-loop poles and/or zeros, that are also on the real axis.

Why? By Equation 12, the angles must add up to an odd multiple of 180° .

- ▶ A complex conjugate pair of open-loop zeros or poles will contribute zero to this angle.
- ▶ An open-loop pole or zero on the real axis, but to the left of the respective point, contributes zero to the angle.
- ▶ The number must be odd, so they add to an odd multiple of 180° , not an even one.

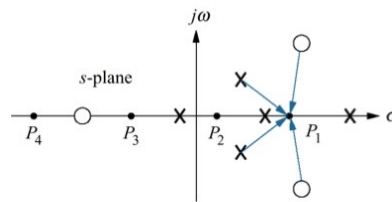


Figure 8.8.

3. Root locus在real axis上只存在于：从右往左数第奇数个pole/zero的左边。只数实数轴的。

对于多次方的pole/zero，如果是奇数次（1，3，5）那么就当作一次方来处理，如果是偶次方（2，4，6），那么该点左侧右侧的root locus状态不会发生变化（都有或者都没有）。

注意：每个pole/zero都会从该点产生root locus曲线，普通pz产生一个，多次方pz产生n个曲线，所以对于多次方pz，其产生的root locus可能直接进入复平面而不是在real axis上。

我们怎么处理无穷远：我们假设无穷远有可能是pole或者zero，但是目的是

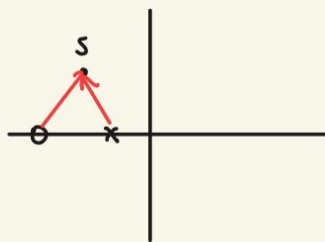
为了让pole数量=zero数量，我们先比较G的pole和zero数量，然后缺的就从无穷上补。注意：如果无穷被等价于zero或者pole，有可能会和系统G的zero/pole产生的root locus相撞。

Why: Note that we want $|KG(s)H(s)|=1$ and $\angle = (2k+1)180^\circ$, $k \in \mathbb{N}$

$$KG(s)H(s) = K \frac{(s-z_1)(s-z_2)\dots}{(s-p_1)(s-p_2)\dots}$$

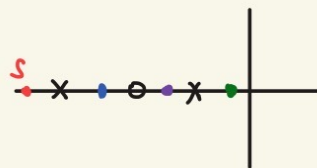
in graph, if we choose randomly

a point s :



vectors $\in (s-z)$ or $(s-p)$

For example:



① When s at $\bullet$

Note: magnitude $\in k$, so we only care about $\angle$

$$\angle = \sum \angle z - \sum \angle p = 180^\circ - (180^\circ \times 2) = -180^\circ = (-2+1)180^\circ$$

so: satisfy the requirement, root locus exists.

② When s at $\bullet$

$$\angle = 180 - (0+180) = 0^\circ, \text{ does not satisfied } \Rightarrow \text{no root locus}$$

③ When s at $\bullet$

$$\angle = 0 - (0+180) = -180^\circ \text{ so has root locus}$$

④ When s at $\bullet$

$\angle = 0 - 0 = 0^\circ$, no root locus

例子:

$$(c) G(s) = \frac{s+7}{(s+8)(s+9)(s+3)^2}$$

poles: $-8, -9, -3$ (2nd order), zero: -7

$$\therefore n=4, m=1$$

$(-\infty, -9)$ 5 p & z yes

$(-9, -8)$ 4 p & z no

$(-8, -7)$ 3 p & z yes

$(-7, -3)$ 2 p & z no.

$(-3, \infty)$ 0 p & z no.

$\therefore$ on $(-\infty, -9)$ & $(-8, -7)$

$n-m=3$, 3 asymptotes

$$\theta_k = \frac{(2k+1)180^\circ}{3}, \quad k=0,1,2$$

$$\theta_k = 60^\circ, 180^\circ, 300^\circ$$

$$\sigma_a = \frac{-8-9-3-3-(-7)}{3}$$

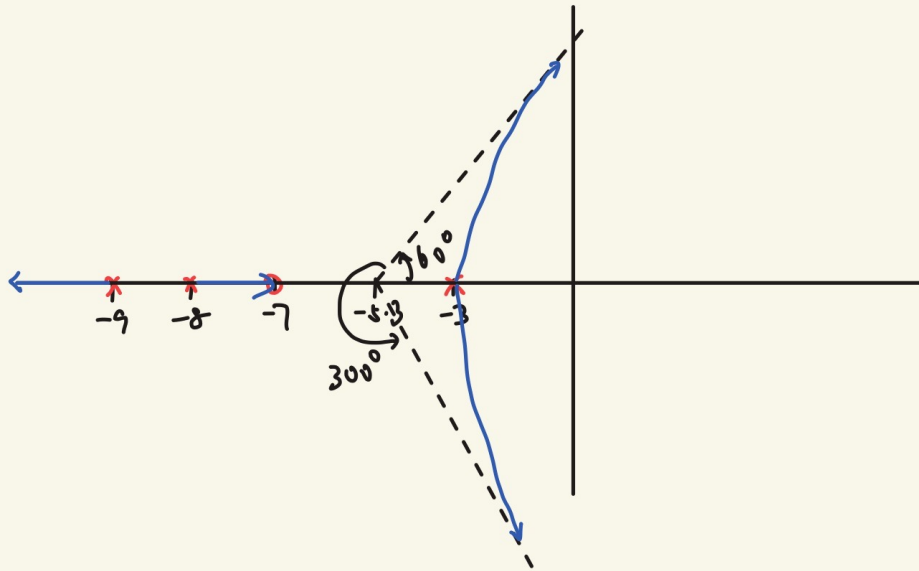
$$= \frac{-23+7}{3} = -5.33$$

$$k = - \frac{(s+8)(s+9)(s+3)^2}{s+7}$$

$$\text{let } \frac{dk}{ds} = 0$$

$$s = -3, -8.46, -6.6 \pm 1.29i$$

choose $s = -3$



Sketching Root Locus - III

4. **Starting and ending points:** The root locus begins at the finite and infinite poles of $G(s)H(s)$ and ends at the finite and infinite zeros of $G(s)H(s)$.

Why? Consider the transfer function below

$$T(s) = \frac{KN_G(s)D_H(s)}{D_G(s)D_H(s) + KN_G(s)N_H(s)}$$

- The root locus begins at zero gain, thus for small K , our denominator is

$$D_G(s)D_H(s) + \epsilon \quad (14)$$

- The root locus ends as K approaches infinity, thus our denominator becomes

$$\epsilon + KN_G(s)N_H(s)$$

4. Root locus曲线从 $G(s)H(s)$ 的poles出发，然后回到 $G(s)H(s)$ 的zeros。

这是因为如果我们把 G 和 H 写成 $G(s) = \frac{N_G(s)}{D_G(s)}$, $H(s) = \frac{N_H(s)}{D_H(s)}$, 然后闭环transfer function就变成ppt中的样子，当root locus刚开始出发， K 接近于0，所以此时pole由 $D_G(s)D_H(s)$ 决定，这是 $G(s)H(s)$ 的pole；而当 K 很大的时候，接近终点，此时闭环传递函数的pole由 $KN_G(s)N_H(s)$ 决定，而这是 $G(s)H(s)$ 的zeros。

Infinite Poles and Zeros

- Consider the open-loop transfer function below

$$KG(s)H(s) = \frac{K}{\epsilon(\epsilon + 1)(\epsilon + 2)} \quad (15)$$

- ▶ From *point 4*, we would expect our three poles to terminate at three zeros, but there are no finite zeros.
- ▶ A function can have an **infinite zero** if the function approaches zero as s approaches infinity. ie. $G(s) = \frac{1}{s}$.
- ▶ A function can have an **infinite pole** if the function approaches infinity as s approaches infinity. ie. $G(s) = s$.
- ▶ When we include infinite poles and zeros, every function has an equal number of poles and zeros

$$\lim_{s \rightarrow \infty} KG(s)H(s) = \lim_{s \rightarrow \infty} \frac{K}{s(s+1)(s+2)} \approx \frac{K}{s \cdot s \cdot s} \quad (16)$$

How do we locate where these zeros at infinity are so we can terminate our root locus?

如果开环传递函数的pole比zero多的话，那么root locus就会发散向无穷远处。

Sketching Root Locus - IV

5. Behavior at Infinity: *As the locus approaches infinity, it approaches straight lines as asymptotes.*

The asymptotes intersect the real-axis at σ_a , and depart at angles θ_a , as follows:

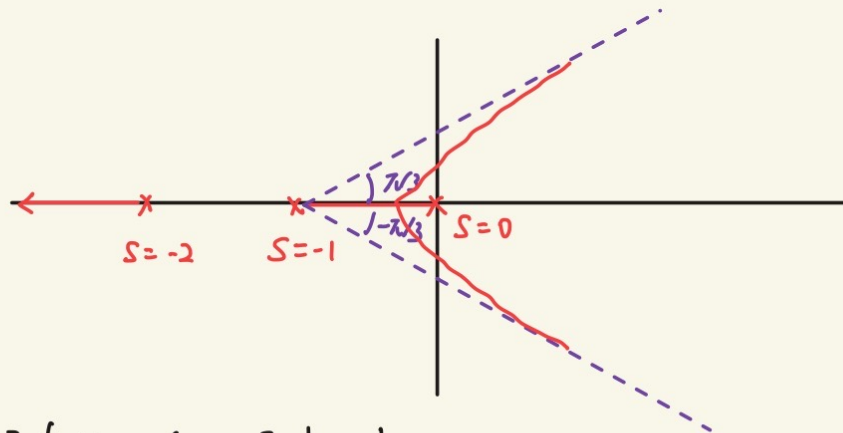
$$\sigma_a = \frac{\Sigma \text{finite poles} - \Sigma \text{finite zeros}}{\# \text{finite poles} - \# \text{finite zeros}} \quad (17)$$

$$\theta_a = \frac{(2k+1)\pi}{\# \text{finite poles} - \# \text{finite zeros}} \quad (18)$$

where $k = 0, \pm 1, \pm 2, \pm 3$, and the angle is in radians relative to the positive real axis.

5. 当root locus发散向无穷的时候，其会按照asymptote发散，而asymptote的交点和离开的角度公式如上：

$$kG(s)H(s) = \frac{k}{s(s+1)(s+2)}$$



step 1: 3 finite poles, 3 branches

step 2: symmetry

step 3: real axis segment

step 4: behavior at infinity $\Rightarrow$ approach asymptote

$$\sigma_a = \frac{0 + (-1) + (-2)}{3} = -1$$

$$\theta_a = \frac{(2k+1)\pi}{3} = \frac{\pi}{3} \text{ for } k=0$$

$$= \pi \text{ for } k=1$$

$$= -\frac{\pi}{3} \text{ for } k=-1$$

Sketching Root Locus eg. 1

- Sketch the root locus for system below.

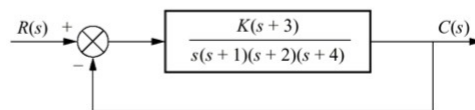
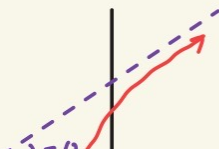
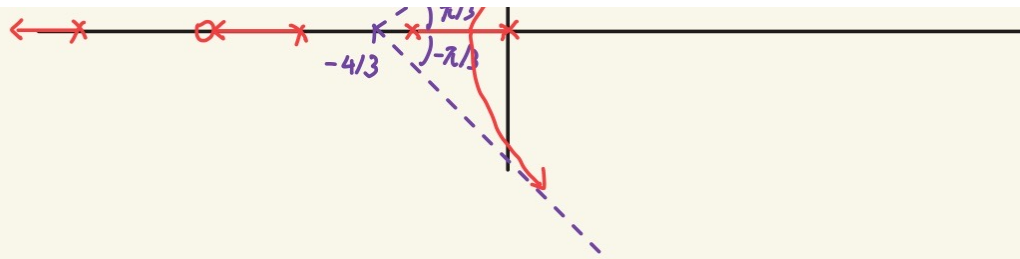


Figure 8.11.





$$\sigma_a = \frac{0 + (-1) + (-2) + (-4) - (-3)}{4 - 1} = -\frac{4}{3}$$

$$\theta_a = \frac{(2k+1)\pi}{4-1} = \frac{\pi}{3} \quad k=0$$

$$= \pi \quad k=1$$

$$= \frac{5\pi}{3} \quad k=2$$

Real-axis Breakaway and Break-in Points

- Consider root locus below.
- We want to be able to calculate at what points on the real axis does the locus leave the real-axis (**breakaway point**), and at what point we return to the real-axis (**break-in point**).
- At breakaway/break-in points, the branches form an angle of $180^\circ/n$ with the real axis where n is number of poles converging on the point.

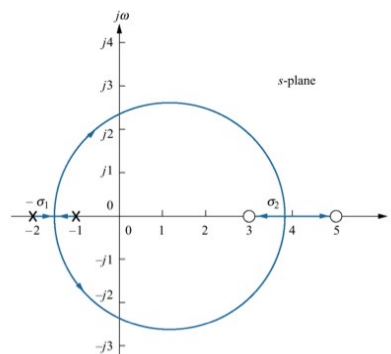


Figure 8.13.

当出现一段区间内多个root locus挤在一起，比如说这个区间的左右端点都是pole或者zero，那么势必会出现breakaway和break in，就是root locus在real axis上分开，进入complex plain。

Real-axis Breakaway and Break-in Points - II

- ▶ Breakaway points occur at maximums in the gain for that part of the real-axis.
- ▶ Break-in points occur at minimums in the gain for that part of the real-axis.
- ▶ We can thus determine the breakaway and break-in points by setting $s = \sigma$, and setting the derivative of equation below equal to zero:

$$K = \frac{-1}{G(\sigma)H(\sigma)} \quad \frac{dK}{d\sigma} = 0 \quad (19)$$

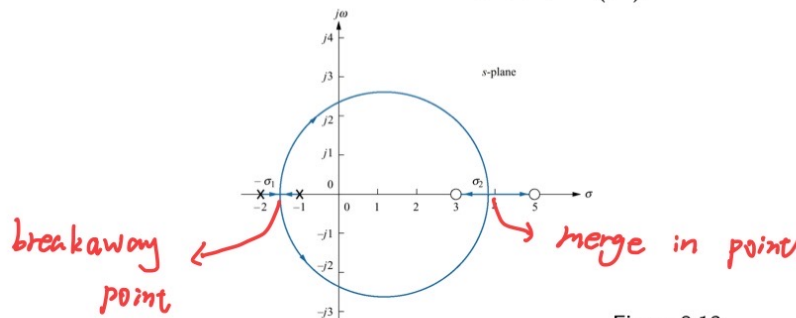


Figure 8.13.

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注意：在root locus，我们只关注 $K > 0$ 的情况，因为这保证了 K 不会改变角度。我们可以把 K 看作关于 σ 的函数，这么做的意义是当一个pole在实数轴上移动的时候， K 会怎么变化（反过来看root locus的意义）。当处于break点的时候， K 对于 σ 的瞬时变化率是0，我们就可以因此求出break点。

算出来的最大点是breakaway，最小点是break in。同时，如果算出来很多结果，那么保留 σ 在real axis和对应的 $K > 0$ 的结果。

$$T(s) = \frac{k G(s)}{1 + k G(s) H(s)} \quad , \quad k G(s) H(s) = -1$$

$$k \left[\frac{(s-3)(s-5)}{(s+1)(s+2)} \right] = -1$$

$$k = \frac{-1}{G(s) H(s)} \quad , \quad k = \frac{-(\sigma+1)(\sigma+2)}{(\sigma-3)(\sigma-5)} \rightarrow \begin{array}{l} \text{max \& min of this} \\ \downarrow \qquad \qquad \downarrow \\ \text{break out} \qquad \text{break in} \end{array}$$

$$\frac{dk}{d\sigma} = \frac{-11\sigma^2 + 26\sigma + 6}{(\sigma-3)^2(\sigma-5)^2} = 0$$

$$\sigma_1 = -1.45, \quad \sigma_2 = 3.82$$

should also calculate break in break out points

Only breakaway points here. Occurs at maximum

$$k = \frac{-1}{G(s) H(s)} \quad , \quad k = -\sigma(\sigma+1)(\sigma+2)$$

$$k = -\sigma^3 - 3\sigma^2 - 2\sigma$$

$$\frac{dk}{d\sigma} = -3\sigma^2 - 6\sigma - 2 \quad , \quad \text{let } \frac{dk}{d\sigma} = 0$$

$$\sigma_1 = -0.42, \quad \sigma_2 = -1.58$$

↓
breakaway point

Real-axis Breakaway and Break-in Points - III

An alternative method for computing the real-axis breakaway and break-in points without differentiation is to solve the equation:

$$\sum_{i=1}^m \frac{1}{\sigma + z_i} = \sum_{i=1}^n \frac{1}{\sigma + p_i} \quad (8.37)$$

where z_i and p_i are the negative of the zeros and poles, respectively, of $G(s)H(s)$.

i.e.

$$G(s)H(s) = \frac{K_{GH}(s + z_1)(s + z_2) \dots (s + z_m)}{(s + p_1)(s + p_2) \dots (s + p_n)}$$

这是一种算break point的替代方案，如果不想算求导就算这个。算出来的 σ 的所有值就是break points。

The $j\omega$ -Axis Crossings

- ▶ For systems like the one below, finding the $j\omega$ -axis crossing is important as it is the value of the gain where the system goes from stable to unstable.
- ▶ Can use the Routh-Hurwitz criteria to find crossing:
 1. Force a row of zeros to get gain
 2. Determine polynomial for row above to get ω , the frequency of

oscillation.

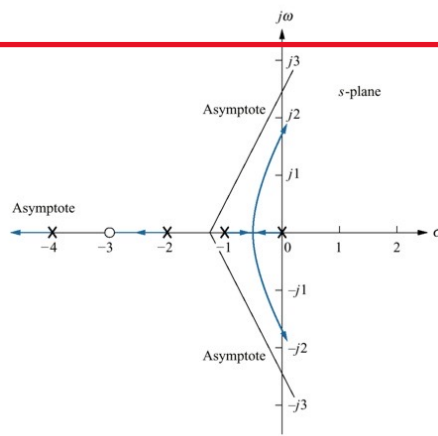


Figure 8.12.

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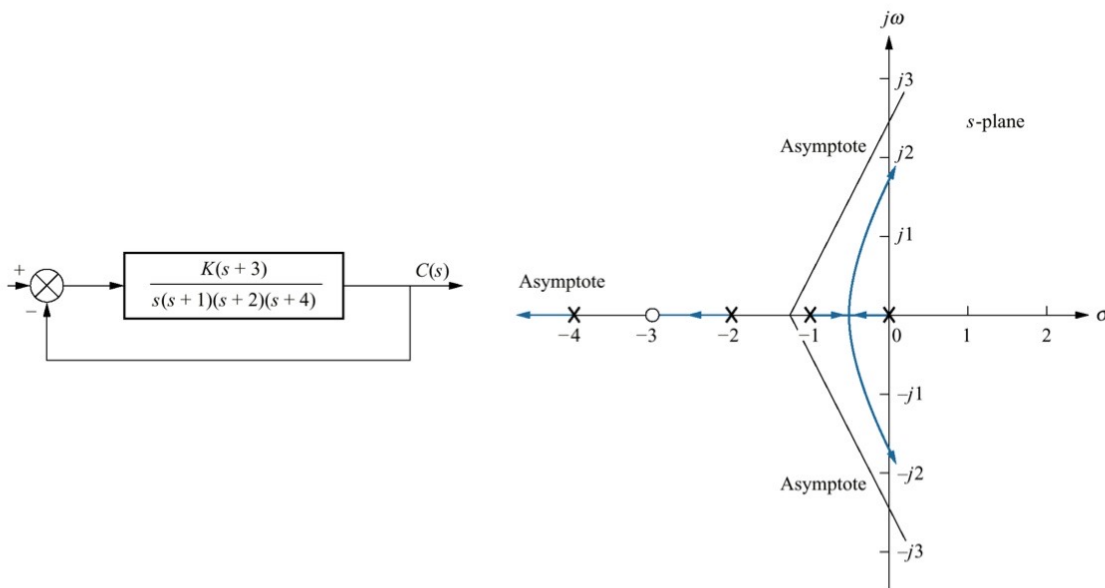
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我们很希望找到root locus穿过虚轴的那个点，因为记住：RHP的pole代表着稳定，RHP是不稳定，所以虚轴上的那个点就是临界值。

因此，我们希望找到和虚轴相交的时候的K以及此时的频率 ω 是多少。

The $j\omega$ -Axis Crossing eg.

- For system below, find the frequency and gain for which the system crosses the $j\omega$ -axis.



Figures 8.11 and 8.12.

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4 poles $\rightarrow$ 4 branches, 1 zero, 3 branches to ∞

$$\sigma_a = \frac{0+(-1)+(-2)+(-4)-(-3)}{4-1} = -\frac{4}{3}, \quad \theta_a = \frac{(2k+1)}{4-1}\pi$$

$$\theta_a = \frac{\pi}{3} \text{ for } k=0, \pi \text{ for } k=1, -\frac{\pi}{3} \text{ for } k=-1$$

Break in / away point: no break in

$$K = \frac{-1}{G(\sigma)H(\sigma)}, \quad K = \frac{-\sigma(\sigma+1)(\sigma+2)(\sigma+4)}{\sigma+3}$$

$$\frac{dK}{d\sigma} = \frac{-(3\sigma^4 + 26\sigma^3 + 77\sigma^2 + 84\sigma + 24)}{(\sigma+3)^2} = 0$$

$$\sigma = -0.43, -1.61, -3.31 \pm 0.68j$$

$\downarrow$
Not on
root locus

$\downarrow$
Not on real axis

Second method: breakaway point

$$\sum_{i=1}^m \frac{1}{\sigma+z_i} = \sum_{i=1}^n \frac{1}{\sigma+p_i}$$

$$\frac{1}{\sigma+3} = \frac{1}{\sigma} + \frac{1}{\sigma+1} + \frac{1}{\sigma+2} + \frac{1}{\sigma+4}$$

$$\sigma = -0.43, -1.61, -3.31 \pm 0.68j$$

The jw crossing

$$G_{CL} = \frac{KG}{1+KGH} = \frac{K(\sigma+3)}{(\sigma+1)(\sigma+2)(\sigma+4)} = \frac{K(\sigma+3)}{1 + K(\sigma+3)}$$
$$= \frac{K(\sigma+3)}{\sigma^4 + 7\sigma^3 + 14\sigma^2 + K\sigma + 3K}$$

$$(s+1)(s+2)(s+4)$$

s^4	1	14	$3k$
s^3	7	$k+8$	0
s^2	$-\frac{\begin{vmatrix} 1 & 14 \\ 7 & k+8 \end{vmatrix}}{7}$ $= \frac{90-k}{7}$	$-\frac{\begin{vmatrix} 1 & 3k \\ 7 & 0 \end{vmatrix}}{7}$ $= 3k$	$-\frac{\begin{vmatrix} 1 & 0 \\ 7 & 0 \end{vmatrix}}{7}$ $= 0$
s^1	$-\frac{\begin{vmatrix} 7 & k+8 \\ \frac{90-k}{7} & 3k \end{vmatrix}}{\frac{90-k}{7}}$ $= \frac{k^2+65k-720}{k-90}$	0	0
s^0	$3k$	0	0

$$-\frac{\begin{vmatrix} 90-k & 3k \\ \frac{k^2+65k-720}{k-90} & 0 \end{vmatrix}}{\frac{k^2+65k-720}{k-90}}$$

Force row in Routh table to be 0

for $k > 0$, can only do this for s row

$$\frac{k^2 + 65k - 720}{k - 90} = 0 \quad k \approx 9.65, -74.6 \text{ (reject)}$$

Method

Use row above row of zeros $\rightarrow$ polynomial

$$\left(\frac{90-k}{7}\right)s^2 + 3k = 0$$

$$(90-k)s^2 + 21k = 0 \quad \text{sub in } k \approx 9.65 \quad s \approx \pm 1.59j$$

Method 2

Setting closed loop denominator to zero for value of k that makes system unstable

$$G_{CL} = \frac{k(s+3)}{s^4 + 7s^3 + 14s^2 + k + 8 + 3k}$$

$s = j\omega \Rightarrow$ on imaginary axis crossing to RHP

$$G_{CL} = \frac{k(j\omega + 3)}{\omega^4 - 7\omega^3j - 14\omega^2 + (k+8)j\omega + 3k}$$

$$\operatorname{Re}(D_{G_{CL}}) = 0$$

$$\omega^4 - 14\omega^2 + 3k = 0$$

$$\omega_R = \pm 1.59, \pm 3.39$$

$$\operatorname{Im}(D_{G_{CL}}) = 0$$

$$-7\omega^3j + (k+8)j\omega = 0$$

$$k \approx 9.65 \Rightarrow \text{Routh Table}$$

$$\omega_L \approx \pm 1.59, 0$$

$\therefore$ choose $\omega = \pm 1.59$ since it makes real & im part be both 0

Angles of Departure and Arrival

- We can refine our sketch by determining at which angles we depart from complex poles and arrive at complex zeros

depart from complex poles, and arrive at complex zeros.

- Net angle from all open-loop poles and zeros to a point on root locus must satisfy:

$$\Sigma \text{zero angles} - \Sigma \text{pole angles} = (2k + 1)180^\circ \quad (20)$$

- To find angle θ_1 , we choose a point ϵ on root locus near complex pole, and assume all angles except θ_1 are to the complex pole instead of ϵ . Can then use Equation 20 to solve for θ_1 .

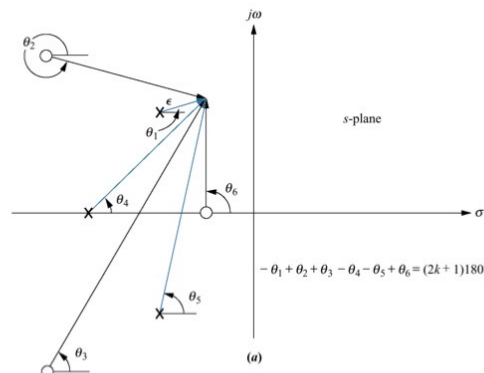


Figure 8.15.

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我们对于在复平面的pole/zero还需要确定其 angle of departure/arrival, 公式的意思是对 root locus 上任一点, 所有 zero 对这个点的角度和, 减去所有 pole 对这个点的角度和, 必须等于奇数倍 180° 。

角度方向的定义是: positive real axis到root locus离开这个点的时候, 切线方向的角度, 角度永远是正实轴到切线方向。想求谁的d/a角就选谁当参考点。

$$G(s)H(s) = \frac{K(s+2)}{(s+3)(s^2+2s+2)}$$

zeros: $s = -2$, poles: $s = -3, -1+j, -1-j$

$$\therefore \Sigma \text{zero angles} - \Sigma \text{pole angles} = (2k+1)180^\circ, k \in \mathbb{N}$$

reference point for angles...

∴ angle for calculating angle is $-1+j$

$$\therefore \theta_3 - (\theta_1 + \theta_4 + \theta_2) = (2k+1)180^\circ$$

$$\therefore \theta_1 = -\theta_2 + \theta_3 - \theta_4 - (2k+1)180^\circ$$

$$= -\tan^{-1}\left(\frac{2}{0}\right) + \tan^{-1}\left(\frac{1}{1}\right) - \tan^{-1}\left(\frac{1}{2}\right) - (2k+1)180^\circ$$

$$\theta_1 = -90^\circ + 45^\circ - 26.57^\circ + 180^\circ$$

$$= 108.43^\circ$$

Angles of Departure and Arrival - II

- For example in Figure 8.15a, we can solve for θ_1 in equation below:

$$\theta_2 + \theta_3 + \theta_6 - (\theta_1 + \theta_4 + \theta_5) = (2k + 1)180^\circ \quad (21)$$

- Similar approach can be used to find angle of arrival of complex zero in figure below.
- Simply solve for θ_2 in Equation 21.

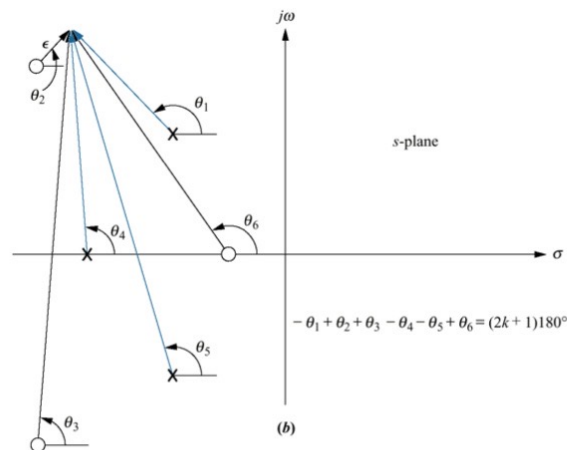
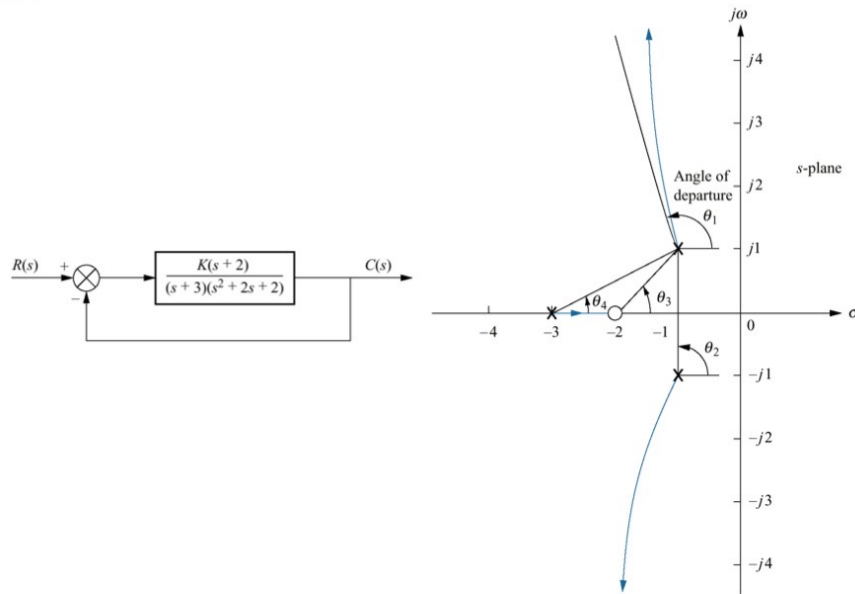


Figure 8.15.

Angles of Departure and Arrival eg.

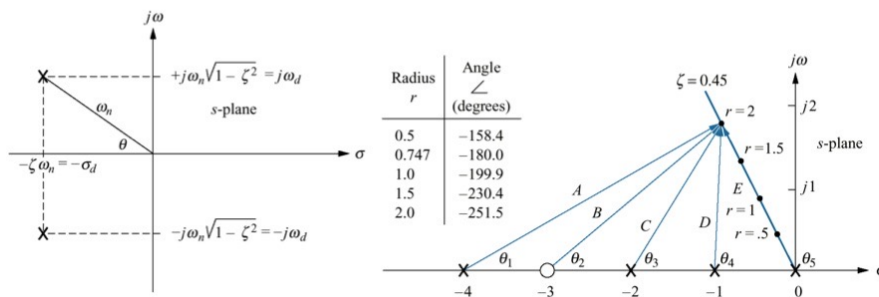
- Find angle of departure for complex poles, and sketch root locus.



Figures 8.16 and 8.17.

Plotting and Calibrating Root Locus

- Once sketched, we may wish to accurately locate certain points and their associated gain.
- For example, we may wish to determine the exact point the locus crosses the 0.45 damping ratio line in figure below.
- From Figure 4.17, we see that $\cos(\theta) = \frac{\text{adj}}{\text{hyp}} = \frac{\zeta\omega_n}{\omega_n} = \zeta$.
- We then use computer program to try sample radiuses, calculate the value of s at that point, and then test if point satisfies angle requirement.



$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\text{poles: } s = \frac{-2\zeta\omega_n \pm \sqrt{(2\zeta\omega_n)^2 - 4\omega_n^2}}{2}$$

$$s = -\zeta\omega_n \pm \sqrt{\zeta^2\omega_n^2 - \omega_n^2}$$

$$= -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1}$$

for underdamp $0 < \zeta < 1$

$$s = -\zeta\omega_n \pm j\omega_n \sqrt{1 - \zeta^2}$$

我们可以通过root locus来找满足我们要求的点；以这里为例子，我们要找 $\zeta = \text{某个数}$ ，我们可以先根据关系求出 θ ，然后画出对应射线，我们要找的就是root locus和这个射线的交点。

我们可以根据等价关系，用zero-pole角度看是不是 $(2k+1)*180$ 来确定是不是在root locus上。然后我们可以找到对应的 r ，radius即 ω_n 的大小。

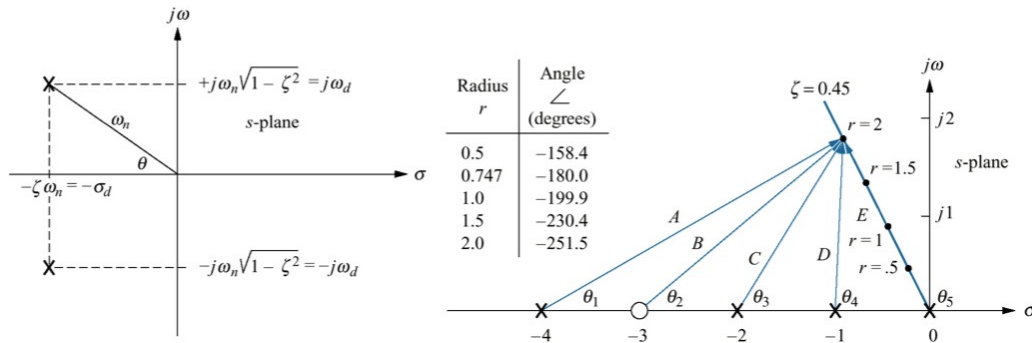
Plotting and Calibrating Root Locus - II

- Once we have found our point we can use the equation below to solve for the required gain, K .

$$K = \frac{1}{|G(s)||H(s)|} = \frac{\prod_{i=1}^n p_i}{\prod_{i=1}^n M_{z_i}} \quad (22)$$

- Uses labels in Figure 8.18, we would have for our example:

$$K = \frac{ACDE}{B} \quad (23)$$



Figures 4.17 and 8.18.

我们也可以顺便把K找出来，公式的意义是K=pole向量的magnitude相乘/zero向量的magnitude相乘。

我们验证一个点是不是在root locus上就是先通过找G在real axis上的pole和zeros，然后以他们为起始点，我们关心的点为终点画vector，然后判断real axis正半轴到该向量的角度，和是不是180的奇数倍。

Transient Response Design via Gain Adjustment

- We want to be able to apply our transient response parameters and equations for second-order underdamped systems to our root locuses.
- These are only accurate for second-order systems with no finite zeros, or systems that can be approximated by them.
- What are the conditions that must be true for a 2nd order approximation to be “close” to the higher order system?

Recall if $G(s) = \frac{N_G(s)}{D_G(s)}$ and $H(s) = \frac{N_H(s)}{D_H(s)}$, closed loop TF is:

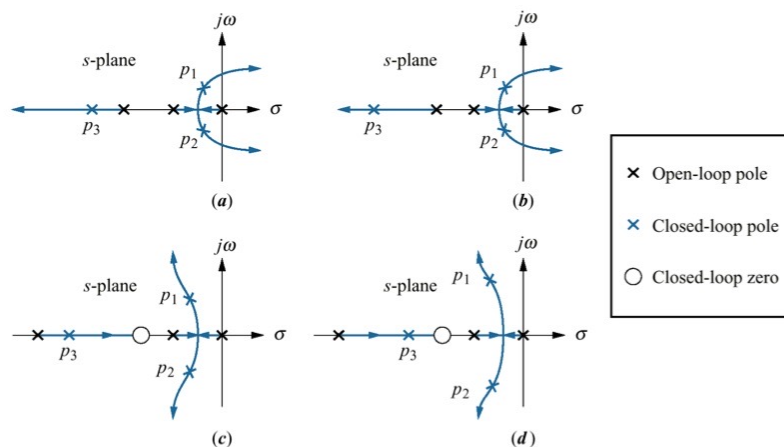
$$T(s) = \frac{K N_G(s) D_H(s)}{D_G(s) D_H(s) + K N_G(s) N_H(s)} \quad (24)$$

我们之前在time response里面计算过那些的公式大部分都只适用于2个pole没有zero的二阶系统，对于高阶系统，我们得看能不能近似。

Transient Response Design via Gain Adjustment II

Conditions for justifying 2nd order approx of a higher order systems

1. Higher order poles are much farther left (e.g. $> 5\times$) of the s -plane dominant closed loop poles. (Holds for (b), not (a))
2. The closed-loop zeros near the two dominant closed-loop poles must be nearly canceled by higher-order poles near them. (Holds for (d), not (c))
3. Closed-loop zeros not cancelled, must be far away from the two dominant closed-loop poles.



几个等效规则：

1. 在高阶系统，超过2个的其他pole比如五倍于dominant pole远，即距离最左边的dominant pole 5倍左远。

2. 如果有zero很靠近dominant poles，他们最好被和他们几乎重合的pole

2. 如果有zero很靠近dominant poles, 他们取对彼此和它们几乎重合的pole抵消掉 (注意, 抵消让zero和pole都失去作用)。
3. 如果zero不能被抵消, 那么它最好离dominant poles远一点。

Defining Parameters on Root Locus

- ▶ We have already seen that as $\zeta = \cos \theta$, vectors from the origin are lines of constant damping ratio.
- ▶ As percent overshoot is solely a function of ζ , these lines are also lines of constant %OS.
- ▶ From diagram we can see that the real part of a pole is $\sigma_d = \zeta \omega_n$, and the imaginary part is $\omega_d = \omega_n \sqrt{1 - \zeta^2}$.
- ▶ As $T_s = \frac{4}{\zeta \omega_n} = \frac{4}{\sigma_d}$, vertical lines have constant values of T_s .

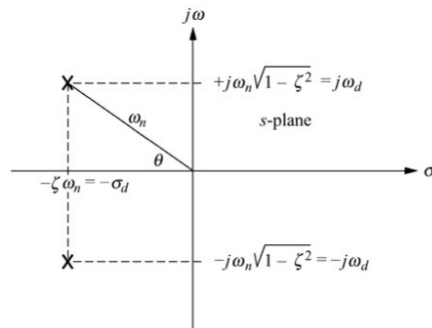


Figure 4.17.

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两个结论:

$\zeta = \cos \theta$, 算出 θ 可以得到射线。

$T_s = 4/\sigma_d$, 算出 $\sigma = \sigma_d$ 这个竖线。

所以, pole必须在射线和real axis夹成的区域里面, 并且必须在竖线的左侧 (因为要满足OS%小于给定值和settling time小于给定值)。

Defining Parameters on Root Locus - II

- ▶ As $T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} = \frac{\pi}{\omega_d}$, horizontal lines thus have

constant peak time.

- We thus choose a line with the desired property, and test to find where it intersects our root locus.

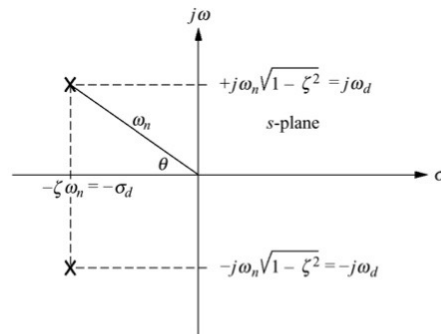


Figure 4.17.

这里是引入peak time的限制， T_p 正好等于 π/ω_d ，其表现为横线限制，注意： $\omega_d = \omega_n \sqrt{1 - \zeta^2}$

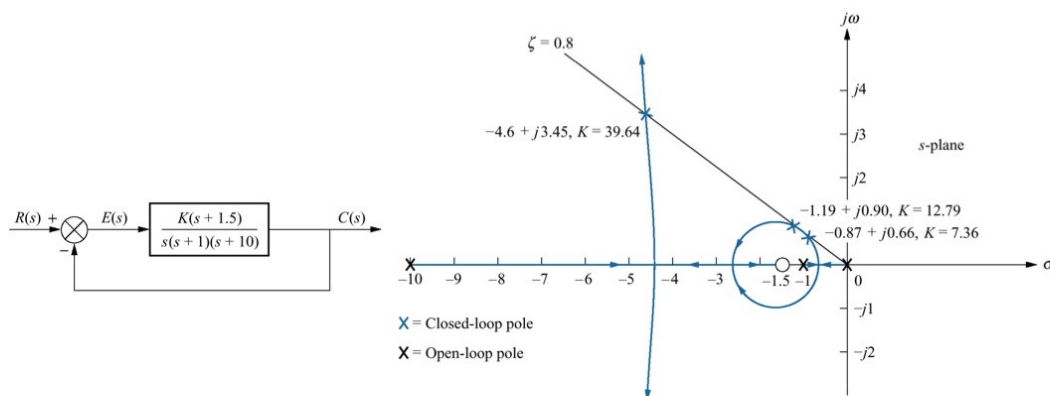
由于我们一般是想让 $T_p < t$ ，所以我们需要让pole在横线夹住的区域以外。

Design Procedure For Higher-order Systems

1. Sketch root locus for system.
2. Assume system has no zeros and is second-order. Find gain that gives desired transient response.
3. Check that systems satisfies criteria to justify our approximation.
4. Simulate system to make sure transient response is acceptable.

Third-order System Gain Design eg.

- ▶ For system below, design the value of gain, K , that will give 1.52% overshoot. Also estimate the settling time, peak time, and steady-state error.
- ▶ First step is to sketch the root locus below.
- ▶ We next assume system can be approximated by second-order system, and solve for ζ using $\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}}$.

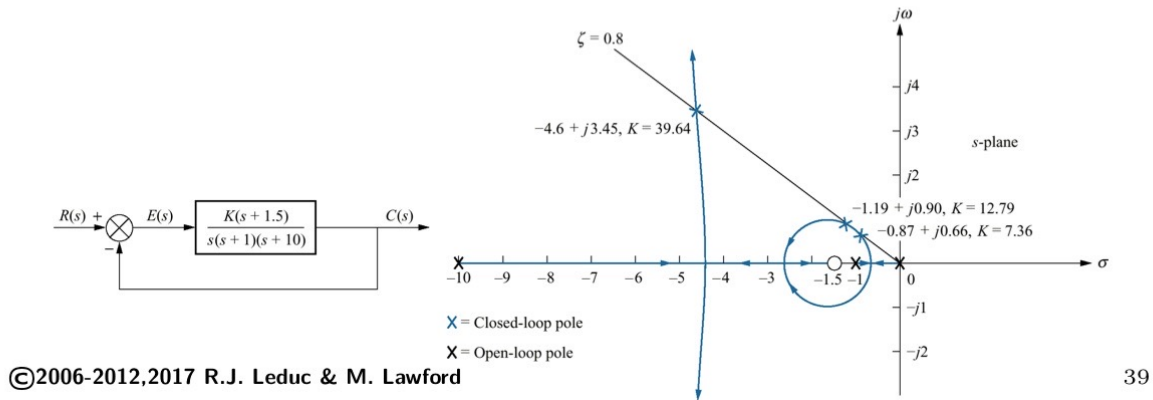


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Third-order System Gain Design eg. - II

- ▶ This gives $\zeta = 0.8$. Our angle is thus $\theta = \cos^{-1}(0.8) = 36.87^\circ$.
- ▶ We then use root locus to search values along this line to see if they satisfy the angle requirement.
- ▶ The program finds three conjugate pairs on the locus and our $\zeta = 0.8$ line. They are $-0.87 \pm j0.66$, $-1.19 \pm j0.90$, $-4.6 \pm j3.45$ with respective gains of $K = 7.36$, 12.79 , and 39.64 .

- We will use $T_p = \frac{\ddot{\cdot}}{\omega_d}$, and $T_s = \frac{\dot{\cdot}}{\sigma_d}$.



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Third-order System Gain Design eg. - III

- For steady-state error, we have:

$$K_v = \lim_{s \rightarrow 0} sG(s) = \lim_{s \rightarrow 0} s \frac{K(s+1.5)}{s(s+1)(s+10)} = \frac{K(1.5)}{(1)(10)} \quad (25)$$

- To test to see if our approximation of a second-order system is valid, we calculate the location of the third pole for each value of K we found.
- The table below shows the results of our calculations.

Case	Closed-loop poles	Closed-loop zero	Gain	Third closed-loop pole	Settling time	Peak time	K_v
1	$-0.87 \pm j0.66$	$-1.5 + j0$	7.36	-9.25	4.60	4.76	1.1
2	$-1.19 \pm j0.90$	$-1.5 + j0$	12.79	-8.61	3.36	3.49	1.9
3	$-4.60 \pm j3.45$	$-1.5 + j0$	39.64	-1.80	0.87	0.91	5.9

Table 8.4.

Third-order System Gain Design eg. - IV

► We now simulate to see how good our result is:

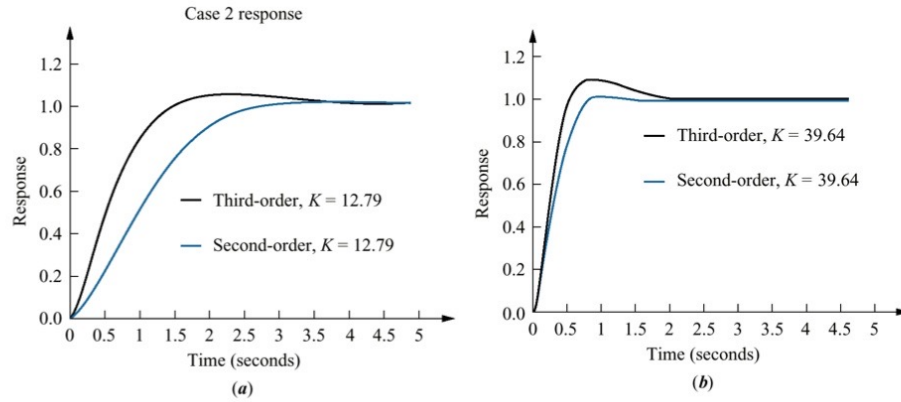
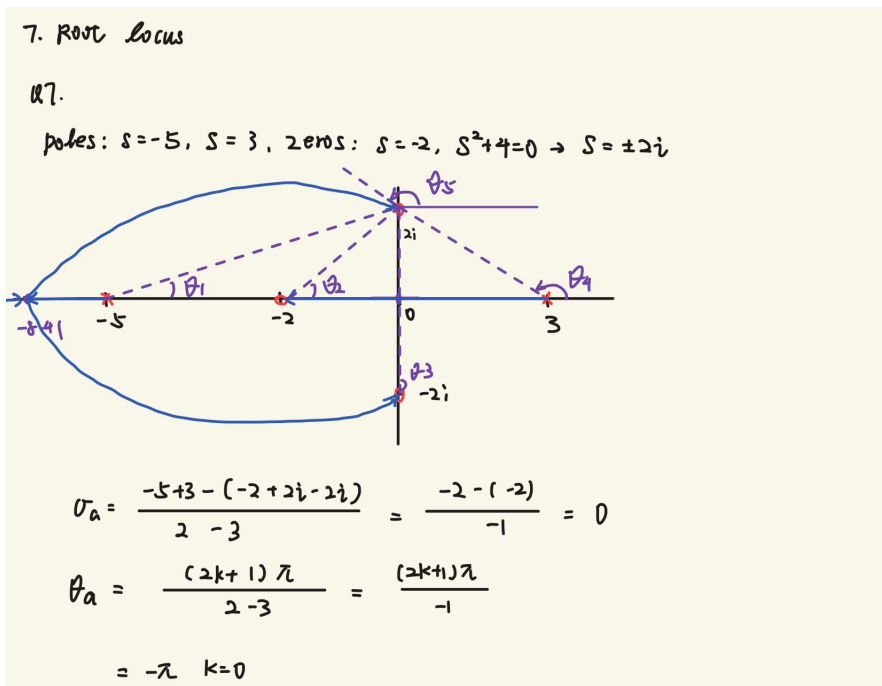


Figure 8.23.

7. Plot the root locus for the unity-feedback system shown in [Figure P8.3](#), where

$$G(s) = \frac{K(s+2)(s^2+4)}{(s+5)(s-3)}$$

For what range of K will the poles be in the right half-plane? [[Section: 8.5](#)]



$$= \pi \quad k = -1$$

$$k = \frac{-1}{G(s)H(s)} = \frac{-1}{\frac{(s+2)(s^2+4)}{(s+5)(s-3)}}$$

$$\frac{dk}{ds} = 0 \quad \therefore \sigma_1 = -0.79 - 0.94i$$

$$\sigma_2 = -0.79 + 0.94i$$

$$\sigma_3 = 5.99$$

$$\sigma_4 = -8.41 \quad \checkmark \text{ break away point}$$

$$K(\sigma_4) = 5.26 \times 10^{-5} > 0$$

intersection of the Im axis:

$$G_{cl} = \frac{KS^3 + 2KS^2 + 4KS + 8K}{KS^3 + (2K+1)S^2 + (4K+2)S + 8K-15}$$

$$\text{let denom} = 0 \quad \text{with } s = j\omega$$

$$\therefore K(j\omega)^3 + (2K+1)(j\omega)^2 + (4K+2)(j\omega) + 8K-15 = 0 + 0j$$

$$\therefore -K\omega^3 j - (2K+1)\omega^2 + (4K+2)j\omega + 8K-15$$

$$\therefore -K\omega^3 + (4K+2)\omega = 0$$

$$-(2K+1)\omega^2 + 8K-15 = 0$$

$\therefore$ To satisfy both equation and make $k > 0$

$$K = 1.875, \quad \omega = 0$$

Angle of arrival:

Calculate angle at $2i$

$$\theta_1 = \tan^{-1}\left(\frac{2}{3}\right) = 21.8^\circ$$

$$\theta_2 = 45^\circ$$

$$\theta_3 = 90^\circ$$

$$\theta_4 = 180 - \tan^{-1}\left(\frac{2}{3}\right) = 146.3^\circ$$

$$\text{let } \theta_2 + \theta_3 - \theta_1 - \theta_4 + \theta_5 = (2K+1)180, \quad K \in \mathbb{N}$$

$$\therefore \theta_5 - 33.1 = (2K+1)180 \quad \text{let } \theta_5 = 180 + 33.1 = 213.1^\circ$$

算系统的性质，比如说 ζ 需要用closed loop $T(s)$ 算，公式单如下：

$$OS\% = e^{-\frac{\zeta * \pi}{\sqrt{1-\zeta^2}}} * 100\%$$

$$\zeta = -\frac{\ln\left(\frac{OS\%}{100}\right)}{\sqrt{\pi^2 + \ln^2\left(\frac{OS\%}{100}\right)}}$$

Settling time:

$$T_s = \frac{4}{\zeta * \omega_n} = \frac{4}{\sigma_d}$$

Peak Time:

$$T_p = \frac{\pi}{\omega_n * \sqrt{1 - \zeta^2}} = \frac{\pi}{\omega_d}$$

确定pole在复平面上的位置： $s = -\sigma \pm \omega_d * j$:

$$\sigma = \zeta * \omega_n = \frac{4}{T_s}$$

ω_d 是阻尼自然频率:

$$\omega_d = \omega_n * \sqrt{1 - \zeta^2} = \frac{\pi}{T_p}$$

ω_n 是自然震荡频率:

$$\omega_n = \sqrt{\sigma^2 + \omega_d^2}$$

阻尼比角度：系统想要满足条件，需要一个dominant pole在阻尼比的放射射线上:

$\theta = \cos^{-1} \zeta$ 极点和负实数轴的夹角

$\phi = 180 - \arccos(\zeta)$ pole在坐标系从正实数轴逆时针算起的角度。

$$\omega_d = \sigma * \tan(180 - \phi)$$

Root locus目标条件:

目标pole s 如果想要在root locus上, 必须满足:

$$\sum \theta_{zeros} - \sum \theta_{poles} = (2 * k + 1) * 180$$

注意: 这 and 把目标点 s_d 带入 $G(s)H(s)$ 求出角度是等价的。

求compensator的补偿角度 ϕ (pole或者zero) :

$$\tan(\phi) = \frac{\text{目标点虚部 } \omega_d}{\text{目标点实部 } \sigma - \text{pole/zero位置}}$$

注意, 计算角度用的是从open loop的zero/pole出发到目标pole点的向量。

第一阶段: 快速画出根轨迹草图 (Sketching)

1. 标出所有的开环极点 (Poles) 和零点 (Zeros)

- **起点与终点:** 根轨迹 (当增益 $K \geq 0$ 时) 总是始于开环传递函数的极点 (此时 $K \rightarrow 0$), 并且终于开环传递函数的零点 (此时 $K \rightarrow \infty$) ¹。
- **分支数量:** 根轨迹的分支 (branch) 数量等于开环极点的数量 ²。

2. 确定实轴上的轨迹 (Real-axis segments)

- 从实轴的最右端向左看: 如果在某一点的右侧, 有限的开环极点和零点的总数是奇数, 那么该点所在的实轴线段就存在于根轨迹上 ²。
- 注意: 复数极点/零点成对出现, 对奇偶性无影响; 如果是多次方 (如2次方、3次方) 的极点或零点, 必须按实际次方数 (个数) 来计算 ¹ ²。

3. 确定对称性 (Symmetry)

- 因为复数极点和零点总是以共轭对的形式出现, 所以根轨迹必定严格关于实轴对称 ²。

4. 画出趋向于无穷远的渐近线 (Behavior at Infinity)

- 如果开环有限极点数 (n) 大于 有限零点数 (m), 那么多出来的 $n - m$ 条分支会发散到无穷远处的“无穷远零点” ¹。它们会沿着特定的渐近线发散。
- 计算渐近线在实轴上的交点 (σ_a):

$$\sigma_a = \frac{\sum \text{有限极点} - \sum \text{有限零点}}{\text{有限极点数} - \text{有限零点数}}$$

- 计算渐近线的发散角度 (θ_a):

$$\theta_a = \frac{(2k + 1)180^\circ}{\text{有限极点数} - \text{有限零点数}} \quad (k = 0, \pm 1, \pm 2 \dots)$$

5. 寻找分离点和汇合点 (Breakaway and Break-in Points)

- 当两条根轨迹在实轴上相向而行并相撞时，它们会离开实轴进入复平面，这就是**分离点 (Breakaway)**；反之从复平面回到实轴并分开的点叫**汇合点 (Break-in)** 3。
- **计算方法 1 (求导法)**: 将增益 K 写成实轴坐标 σ 的函数 $K = \frac{-1}{G(\sigma)H(\sigma)}$ 。令导数 $\frac{dK}{d\sigma} = 0$ 求解 3。分离点对应 K 的局部最大值，汇合点对应局部最小值 3。
- **计算方法 2 (代数法)**: 求解方程 $\sum \frac{1}{\sigma + z_i} = \sum \frac{1}{\sigma + p_i}$ (其中 z_i 和 p_i 是零点和极点的相反数)，解出的 σ 即为可能的突破点 3。

6. 寻找与虚轴的交点 ($j\omega$ -Axis Crossings)

- 虚轴交点代表系统从稳定跨向不稳定的临界点 3。
- **计算方法 1 (Routh-Hurwitz 表)**: 写出闭环特征方程，构建 Routh 表。强制让某一行（通常是 s^1 行）全为 0，解出导致临界稳定的增益 K 值。然后利用全零行上一行的辅助多项式 (Even polynomial) 解出此时的振荡频率 ω 3 4。
- **计算方法 2 (直接代入法)**: 将闭环特征方程的分母强行代入 $s = j\omega$ ，然后令等式的实部和虚部分别等于 0，联立方程组解出 ω 和 K 5。

7. 确定复数极点的出发角与复数零点的到达角 (Angles of Departure and Arrival)

- 如果系统有复数极点或零点，需要精确知道轨迹离开或进入它们的初始切线角度 5。
- **计算依据**: 利用相角条件 $\sum \text{零点相角} - \sum \text{极点相角} = (2k + 1)180^\circ$ 5。
- **具体做法**: 以要求出发角的那个复数极点为“参考点”。计算所有其他开环零、极点到达该参考点的向量角度，代入上述公式，将未知的出发角设为 θ_1 ，即可解出该角度 5。到达角的求法同理。